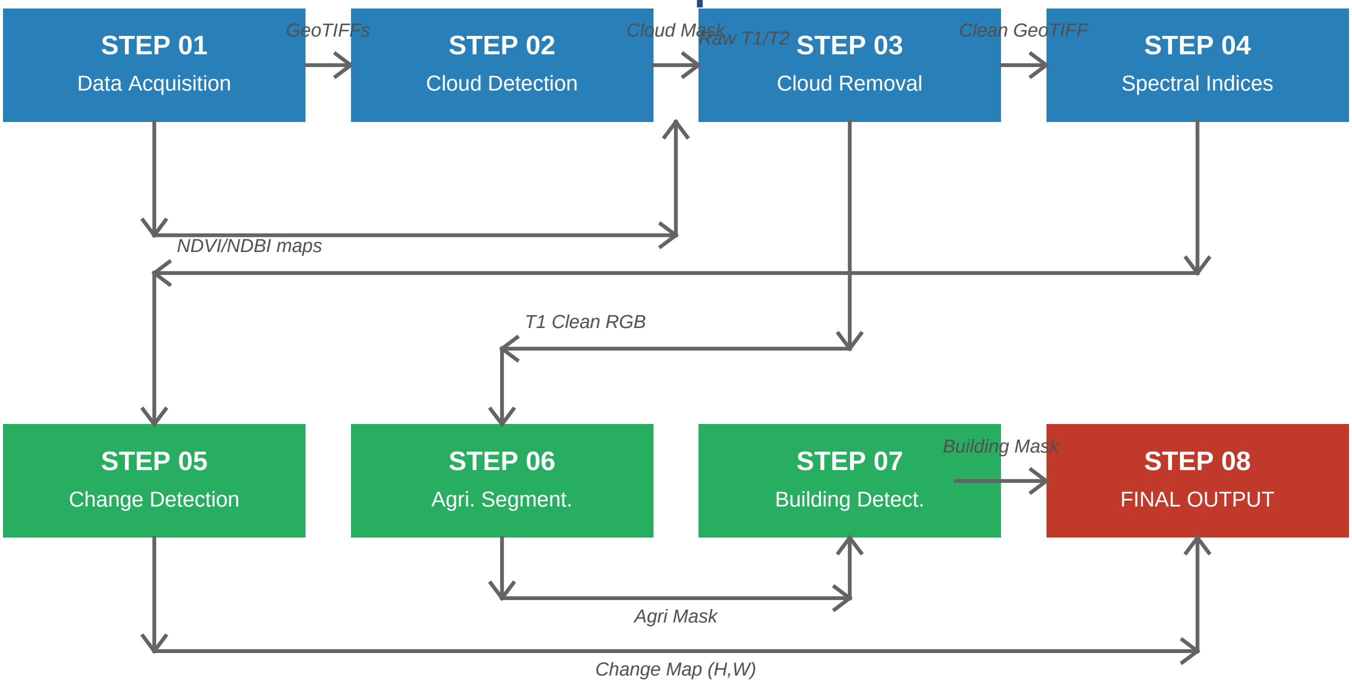
Food Security AI System

Complete Pipeline Flow & Project Details

End-to-End Pipeline Flowchart



1. The One Dataset Per Step

Step	Name	Dataset Used	Pretrained Model
01	Data Acquisition	Sentinel-2 L2A (from GEE)	Google Earth Engine API
02	Cloud Detection	38-Cloud (Landsat clouds)	U-Net + ResNet34 (ImageNet)
03	Cloud Removal	No dataset (uses Step 02 mask)	OpenCV Telea Inpainting
04	Spectral Indices	No dataset (pure formulas)	NumPy Math (NDVI, NDBI)
05	Change Detection	LEVIR-CD (Change pairs)	ChangeFormer (Siamese)
06	Agriculture Segment	ADE20K (150 classes)	SegFormer-B4 (from HF)
07	Building Detection	SpaceNet v2 (Building footprints)	SAM (Segment Anything ViT-B)
08	Final Output	No dataset (uses previous masks)	OpenCV + GeoPandas

2. Detailed Step Specifications & Expected Outputs

Step 01 - Data Acquisition

Goal: Download multi-temporal satellite images (T1=before, T2=after).
Dataset: Sentinel-2 L2A via Google Earth Engine.
Expected Output: Two Multi-band GeoTIFFs (T1 and T2) at 10m resolution.

Step 02 - Cloud Detection

Goal: Detect which pixels are covered by clouds.
Model: U-Net with ResNet34 encoder pretrained on ImageNet.
Dataset: Fine-tunable via 38-Cloud dataset.
Expected Output: Binary cloud mask (H,W) highlighting cloud coverage.

Step 03 - Cloud Removal

Goal: Reconstruct the surface below the clouds.
Model: OpenCV Telea inpainting algorithm.
Expected Output: Cloud-free T1 and T2 GeoTIFFs across all bands.

Step 04 - Spectral Indices

Goal: Compute vegetation (NDVI) and built-up (NDBI) indices.
Model: Pure mathematical band combination via NumPy.
Expected Output: Float32 arrays for NDVI, NDBI for the models directly.

Step 05 - Change Detection

Goal: Identify pixels that changed between the two time periods.
Model: ChangeFormer (Siamese Transformer).
Dataset: Pretrained on LEVIR-CD.
Expected Output: Binary change map where 1 = changed, 0 = unchanged.

Step 06 - Agriculture Segmentation

Goal: Extract pixels classified as agricultural land in the Before (T1) image.
Model: SegFormer-B4.
Dataset: Pretrained on ADE20K (filtering class IDs 9, 29, 92, etc.).
Expected Output: Binary agricultural mask (1 = farmland).

Step 07 - Building Detection

Goal: Segment instances of buildings exclusively within the changed agricultural regions.
Model: SAM (Segment Anything Model by Meta AI) + YOLOv8-seg.
Dataset: Pretrained on SA-1B, optional SpaceNet v2 fine-tuning.
Expected Output: Building masks and polygon bounding coordinates.

Step 08 - Final Output

Goal: Synthesize all masks into an overarching visualization and report.
Color Logic: RED (buildings on farmland), YELLOW (changed vegetation), GREEN (stable agriculture).
Expected Output: High-resolution colored GeoTIFF, PNG preview, GeoJSON polygons, and a JSON area statistics report.

KEMET1 BeforeAfter RF Classifier - Results

The KEMET1 BeforeAfter classifier is a Random Forest trained on 300 co-registered Sentinel-2 tile pairs (before: 2024, after: 2025) covering the Nile Delta. It detects agricultural-land encroachment by analysing 46 spectral-change features derived from NDVI, NDBI, MNDWI and five additional spectral bands.

Model Performance

Dataset:	300 sites - 78 positive / 222 negative	
	70 % train / 15 % val / 15 % test (stratified)	**Split:**
	Random Forest - 400 trees, max_depth=6, balanced weights	**Algorithm:**
	46 (42 Delta-spectral statistics + dNDVI, dNDBI, pct_conv, pct_new)	**Features:**
	0.9444 <- primary / conservative metric	**Val AUC *:**
	~1.000 (inflated: label-feature circularity, see note)	**Test AUC:**
	TN=29 FP=4 FN=2 TP=10 (@ threshold 0.40)	**Val Confusio**
	TN=34 FP=0 FN=0 TP=11 (@ threshold 0.40)	**Test Confusi**
	0.40 (optimised for high recall on agricultural encroachment)	**Decision Thr**

** Note on label-feature circularity: auto-labels were generated using pct_conv and max_cluster_ha thresholds - both of which are also included in the feature vector. The model partially learns to replicate the labelling rule, inflating test AUC toward 1.0. Val AUC = 0.9596 is the honest generalisation estimate (val samples excluded during training). Ablation removing the two circular features (pct_conv, pct_new) yields val AUC = 0.8763 - the conservative lower bound on true generalisation performance.*

Inference Output

run_inference.py generates a self-contained HTML report for each site containing:

- Before / After false-colour composites with change bounding boxes
- Interactive Leaflet map with all detected change clusters
- Reverse-geocoded location names for each cluster
- Area metrics (km2) and RF confidence score

batch_report.py processes all positive sites and builds encroachment_summary.html - a master Egypt map showing all detected sites with their total area lost.